

Supplementary Material

Deep Learning for Leaf Disease Recognition in Rice and Maize: A Systematic Review of Vision Tasks and an Audit of Data Leakage in Model Evaluation

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This document provides methodological details supplementing Section 2 of the main paper: the contribution of each search source (Section S1), the screening process (S2), the amendments to the protocol (S3), the protocol assessment rules (S4), the rules for selecting representative studies (S5), the statistical methods (S6), and the specification of the controlled experiment (S7). The data and logs are archived on Zenodo, <https://doi.org/10.5281/zenodo.23151397>.

S1 Search strategy

The query consists of three blocks combined with AND: crop, leaf disease (including generic terms and the names of the major diseases), and method (including the deep network families for the four tasks CLS, DET, SEG, and GEN). IEEE Xplore uses the same structure as Scopus; the verbatim search strings, search fields, filters, run dates, and number of results for each source are given in Table 2 of the main paper. The contribution of each source to the set of 202 included studies is shown in Table S1: Scopus contains 189 of the 202 studies, 149 of which are found only in Scopus; ScienceDirect contributes 10 unique studies even though its search precision is only 4.0%. Overall search precision was 19.5%, that is, about five records were screened, on average, for each included study [1].

Table S1: Contribution of each database to the set of 202 included studies.

Database	Records	Included studies	Rec _s (%)	P _s (%)	Unique to this source
Scopus	499	189	93.6	37.9	149
IEEE Xplore	41	18	8.9	43.9	2
ScienceDirect	774	31	15.3	4.0	10
MDPI	19	4	2.0	21.1	1

Records are counted before deduplication; a study may appear in more than one database. Rec_s: proportion of included studies found in source *s*; P_s: proportion of records from source *s* that were included [2].

S2 Screening process

Duplicate records were identified by normalized DOI, or by normalized title when the DOI was missing [3]; all 296 duplicates were identified by DOI. In the first screening round, predefined rules applied to the metadata, titles, and abstracts excluded 323 records clearly out of scope and 201 records with an ineligible publication type (96 not published in fully open-access journals, 93 conference proceedings, and 12 reviews). One further review was detected in the second round, bringing the number of records excluded by publication type to 202, as in Figure 1 of the main paper. In the second round, an AI tool (Google Gemini) provided preliminary screening recommendations for the remaining 513 records based on titles and abstracts; one author (T.-H.T.-L.) reviewed each suggestion, another author (M.-H.L.) checked the excluded records, and disagreements were resolved by a third author (T.-N.D.). In total, 168 records were excluded in this round: 95 records with an ineligible research objective according to the second-round decision and 73 records excluded under amendments A and B (Table S2). At the full-text stage, gate (2) excluded studies that used frozen pretrained networks only as fixed feature extractors for classical classifiers (e.g., SVM, *k*-NN, or an extreme learning machine) without training any network layer, or shallow networks on handcrafted features; gate (3) excluded reports with clear data leakage, a test set identical to the training set, no held-out set, or fewer than 200 original images of unclear provenance. Each excluded report was assigned a single code, namely the first gate it failed. The decision and reason for every record are available in the data repository.

S3 Amendments to the protocol

The protocol was registered on the Open Science Framework (OSF) Registries (DOI: 10.17605/OSF.IO/FWZ4S) on 2026-10-03, after the search (2026-09-21–22) and screening had begun. Table S2 lists the amendments in chronological order. The two scope-narrowing amendments (A and B) were decided after some of the full texts had been read, so they may have been influenced by the reports already examined; they were reapplied to all records, and the 73 records excluded under them at title and abstract screening (plus 13 full-text reports under amendment A) are recorded separately. The full protocol with its 12 numbered amendments is available in the data repository; Table S2 summarizes the amendments that affected the set of studies or the results.

Table S2: Amendments to the protocol registered on OSF.

Date	Amendment	Rationale and consequence
2026-09-26	Open-access status of journals verified against the Directory of Open Access Journals (DOAJ) list exported on 2026-09-26 instead of manual lookup	Standardizes the journal criterion; changes for individual journals are logged
2026-09-26	Exclusion codes X1 and X3 at the title and abstract stage applied only when the abstract explicitly states the exclusion reason	Reduces erroneous exclusions; uncertain cases move to full text
2026-09-26	A second title and abstract screening round added, with AI-tool suggestions reviewed by two authors	The rule-based round could not adequately distinguish research objectives; 95 records excluded in this round (168 together with the 73 records under amendments A and B)
2026-09-26	Amendment A: only studies whose experimental data are limited to rice and/or maize included; studies that also use other crops excluded even when they report separate results	Separate results are usually insufficient for coding data and protocol; 41 records excluded at title and abstract screening and 13 reports at full text
2026-09-26	Amendment B: only RGB images included; hyperspectral, multispectral, UAV remote-sensing, and thermal images excluded	Differences in equipment and task; 32 records excluded
2026-09-26	Full-text assessment by four gates instead of keyword checking	The evaluation protocol needs to be appraised at the full-text level
2026-10-03	Leakage labels of 92 studies re-examined against six signs, with the no-separate-validation-set flag separated	The initial labels had no operational criteria; 38 studies relabeled, 5 of them in the representative set (two of which moved to suspected leakage)
2026-10-03/ 2026-10-04	Additional analyses after internal critical review: headline accuracy of the reports excluded for leakage and same-source/cross-source metric pairs of the studies with an external test extracted from the full texts; accuracy compared between protocol groups (Hodges–Lehmann shift with bootstrap confidence intervals and a ± 1 -point equivalence margin, within-source and trend tests, rank regressions, and a citation model); the controlled experiment of Section 7 of the main paper added	Separates the effect of the evaluation protocol from dataset difficulty; descriptive and sensitivity analyses only; the eligibility criteria and the included set are unchanged
2026-10-04	PRE-AUG labels for public datasets assigned from directly measured duplicate-image rates instead of inference from descriptions	The measurements do not support the claim that Kaggle Corn or Maize [4] contains pre-augmented copies; 18 studies relabeled

Table S2 (continued)

Date	Amendment	Rationale and consequence
2026-10-05	Gate (3) applied consistently: reports without a held-out set (NO-HOLDOUT) excluded; TEST-AUG recorded only as a flag, not counted as leakage; one report previously excluded for leakage reread	Following an independent review; 3 reports excluded, 1 report re-included with a suspected-leakage label, 2 studies moved to clean; the included set now has 202 studies

Discrepancies between the OSF registration and the conduct of the review: The retrospective registration (2026-10-03) was drafted before the review was completed, and some descriptions do not match the procedures ultimately reported in the manuscript: (i) the registration states that full-text assessment was performed independently with Cohen's kappa above 0.85, but in this review each full-text report and each label was assessed by one author, with a second author checking every decision, without independent dual assessment or an agreement coefficient; (ii) the registration gives 111 studies (52 rice, 59 maize) in one place and 204 studies in another, which reflect interim counts, whereas the final set comprises 202 studies (Figure 1); (iii) the registration states that the code is publicly stored on GitHub, but the accompanying data repository contains only data, and the analysis code is available upon request; (iv) the exclusion codes E1–E6 in the registration were replaced by the four full-text assessment gates and the exclusion codes in Figure 1 and Section S2; (v) regarding missing data, field 376-56 of the registration correctly states that authors were not contacted, but field 376-58 describes the synthesis of the results of contacting authors; in practice, the authors used only the full text and published appendices of each study, did not contact the original authors, and information that was not reported was coded as not determinable. On 2026-10-07 the registration was updated on OSF; the update has been approved and made public. The rationale for the update lists all discrepancies (i)–(v) and the removal of 296 duplicates by DOI; the description field now gives 202 studies, and the field on full-text assessment now states that one author assessed each report and another author rechecked it, with no agreement coefficient computed. Some other fields still retain the content of the original registration, such as the counts of 111 and 204 studies, the exclusion codes E1–E6, and contact with study authors. For these fields, the update rationale, the main paper, and the Supplementary Material describe the procedures actually followed; where there is a discrepancy, the description here takes precedence. Table S3 maps the exclusion codes E1–E6 of the registration to the actual stages and exclusion codes.

Table S3: Exclusion codes E1–E6 in the OSF registration and the corresponding stages and exclusion codes in the review.

Code in the registration	Actual stage and exclusion code (number of records or reports)
E1, wrong crop	Title and abstract screening: X1 not limited to rice and/or maize (351, including 41 multi-crop records); full text: gate (1)
E2, not deep learning	Screening: X3 (44); full text: gate (2)
E3, not images	Screening: X8 non-RGB images (32); full text: gate (1)
E4, not an empirical study	Screening: X6 reviews (13)
E5, no full text	Not a gate: report could not be retrieved (5)
E6, duplicate	Not a gate: deduplication by DOI before screening (296)
No corresponding code	Screening: not in a fully open-access journal (96), conference proceedings (93), X2 not a leaf disease (59), X4 no quantitative evaluation (4); full text: gate (3) invalid evaluation protocol, including 93 reports with augmentation before splitting, and gate (4)

S4 Protocol assessment rules

The condition for an independent split is Equation (2) of the main paper, where $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{test}}$ are the training and test image sets, $T(x)$ is image x after augmentation T , and $\mathcal{D}_{\text{train}}^{\text{aug}}$ is the training set after augmentation. A violation is considered clear when the text states that augmentation was performed before splitting or when the image counts of the subsets add up to the total number of images after augmentation; when a sign of such a violation is present but the description is insufficient for a definite judgment, the study is included and coded as suspected leakage; when the protocol is described too vaguely to identify either a sign or a clean split, the study is coded as not determinable. The protocol of each study was assessed by one author, and another author rechecked every label. For public datasets, a common rule for all studies using the same dataset is applied only when the duplicate-image rate measured on that dataset is high (Table 5 of the main paper); at present it applies only to the Sethy dataset [5], with the SUB-UNIT sign. Table S4 maps the six signs to the eight-type leakage taxonomy of Kapoor and Narayanan [6]; image augmentation and datasets containing pre-augmented copies are two leakage forms specific to this field.

Table S4: Mapping of the six leakage signs to the taxonomy of Kapoor and Narayanan [6].

Sign in this review	Corresponding type in Kapoor and Narayanan [6]
AUG-FIRST: augmentation or image generation before splitting	L1.2 preprocessing on both the training and test sets
SUB-UNIT: splitting at a unit smaller than the original image, leaf or plant	L3.2 non-independence between training and test samples
PRE-AUG: dataset already contains augmented copies or duplicates	L1.4 duplicates in the dataset
TEST-AUG: test set contains augmented copies of the test images themselves	No corresponding type; does not bring training information into the test set but biases the estimate; recorded only as a flag
NO-HOLDOUT: no held-out set	L1.1 no test set
DUP-LEFT: duplicate images across subsets not handled	L1.4 duplicates in the dataset
No-separate-validation-set flag	Close to L1.3 if the test set is used for selection

The labels of the 92 studies flagged for re-examination were re-examined with assistance from an AI tool, and one author confirmed each label. After the duplicate-image rates were measured directly, the common PRE-AUG rule for Kaggle Corn or Maize [4] was dropped because the measurements did not support it, and 18 studies were relabeled according to their own protocols. The independent review of 2026-10-05 led to three changes: three reports coded NO-HOLDOUT [7–9] were excluded at gate (3) to apply the held-out-set criterion consistently; one report previously excluded for clear leakage [10] was re-included with a suspected-leakage label, because the article does not state the number of original images and does not make clear that augmentation was performed before splitting; and TEST-AUG became a separate flag, so two studies that had only TEST-AUG and split before augmentation were classified as clean.

S5 Selection of representative studies

The results section reports in detail only a set of representative studies, so that the descriptive tables remain readable and studies with weak protocols are not presented on an equal footing with studies with clean protocols. The representative set comprises 38 studies, selected by quota for each task group and crop; the quotas and the selection rationale for each group are summarized in Figure 2 of the main paper. Within each group, studies with clean protocols were considered first; half of the slots (rounded down) were assigned to the clean studies with the most citations per year [11],

$$\text{cpy}_i = \frac{c_i}{\max(1, 2027 - y_i)}, \quad (\text{S1})$$

where c_i is the Scopus citation count on the search date, y_i is the publication year, and $2027 - y_i$ is the number of years in which the study could have been cited, including the search year 2026; the remaining clean studies were then scanned from the newest publication year (ties broken by citations per year), and the first study of each architecture family not yet represented in the group was added; slots still open after this pass (one each for rice detection and maize segmentation) went to the next clean studies by citations per year. Only the image generation group lacked enough clean studies at the time of selection; after the two re-examinations of leakage labels, two selected classification studies moved to the suspected-leakage group and were retained to preserve the original selection rule, so 34 of the 38 representative studies have clean protocols. Citation counts were not an eligibility criterion; they were used only to rank clean studies within a group when filling representative slots. The two single-dataset comparison tables were built following George et al. [12]: each table includes every classification study on that dataset that was classified as clean before the label re-examination and reports accuracy or F1, including studies outside the representative set; studies that moved to the suspected-leakage group after the re-examination are kept in the table and marked with their sign codes.

S6 Statistical methods

Each proportion $\hat{p} = x/n$, where x is the number of studies with the characteristic of interest among n studies, is reported with a 95% Wilson confidence interval [13, 14]:

$$\frac{\hat{p} + \frac{z^2}{2n} \pm z \sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}, \quad z = 1.96, \quad (\text{S2})$$

where z is the 97.5% quantile of the standard normal distribution. Medians and quartiles follow the type 7 sample quantile definition of Hyndman and Fan [15]. Two groups are compared with the two-sided Mann–Whitney test [16], with the rank-biserial correlation as the effect size [17]:

$$U = \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} \left[\mathbf{1}(a_i > b_j) + \frac{1}{2} \mathbf{1}(a_i = b_j) \right], \quad r_{\text{rb}} = 1 - \frac{2U}{n_1 n_2}, \quad (\text{S3})$$

where a_i and b_j are the values of the two groups of sizes n_1 and n_2 , and $\mathbf{1}(\cdot)$ equals 1 when the condition holds and 0 otherwise, so U counts the pairs in which the value in group a exceeds that in group b , with ties counted as one half; r_{rb} lies in $[-1, 1]$, and $r_{\text{rb}} > 0$ means that group b tends to be higher. In the main paper, a is the clean-protocol group and b is the suspected-leakage group. The shift between the two groups is estimated with the Hodges–Lehmann estimator, i.e., the median of all pairwise differences, with a bootstrap confidence interval from 10 000 iterations (2 000 iterations in the sensitivity analysis without the SETH dataset); in an additional analysis added after internal review (OSF amendment, October 2026; not prespecified), the two groups would be considered equivalent if the 90% confidence interval lay within a margin of ± 1 percentage point; the interval (0.55–1.59 points) did not, so equivalence was not shown.

The primary test for the hypothesis that leakage is associated with higher reported accuracy is the two-sided Mann–Whitney test between the suspected-leakage group and the clean group. Because leakage labels were assigned by a common rule for some datasets, three sensitivity analyses for the same hypothesis were corrected for multiple comparisons with the Holm method [18]: a comparison within each data source that contains both groups, using the van Elteren stratified Wilcoxon test; a linear regression of accuracy ranks on group, adjusted for the number of classes, publication year, and the no-separate-validation-set flag, with data-source fixed effects (the model without them is reported descriptively, with an unadjusted p value), using HC3 robust standard errors; and a comparison of the proportion of results of 99.5% or higher using Fisher’s exact test. The comparison of the three groups (clean, suspected leakage, and excluded for clear leakage) using the Jonckheere–Terpstra trend test is an exploratory analysis, reported with two-sided p values; the accuracy of the excluded reports was extracted from the full texts using the same extraction form. Citation counts were modeled by negative binomial regression on group, with publication-year fixed effects and standard errors clustered by journal. All computations were performed in Python with SciPy and statsmodels; the analysis code is available from the authors upon reasonable request.

S7 Specification of the controlled experiment

The training code is not released; this section describes the experimental procedure so that readers can interpret the numerical results in Tables 14 and 15 of the main paper and cross-check them against the files in the data repository.

Data and image groups: Each image is decoded once, resized to 256×256 and retained unchanged across runs. Near-duplicate groups are taken from the group file of each dataset: the broad criterion (`images.csv`) for the SETHY dataset, the PlantVillage maize subset [19], and Kaggle Corn or Maize; the criterion requiring both signals to match (`images_verified.csv`) for Paddy Doctor [20]. The SHA-256 hashes of these files are given in `RUN_COMMANDS.md`.

Splitting by seed: For seed $s \in \{0, \dots, 4\}$, the list of split keys (image groups in GROUP-SPLIT, MATCH-CTRL, and MATCH-LEAK; individual images in RAND-SPLIT; each original image and each augmented copy in AUG-SPLIT) is sorted and shuffled with a random number generator initialized with s ; the first 70% of the keys are assigned to the training set, the next 10% to the validation set and the remainder to the test set. The proportions are computed over keys, not images, so for datasets with large groups the sizes of the three sets deviate slightly from 70/10/20. The same s yields the same split in GROUP-SPLIT, MATCH-CTRL, and MATCH-LEAK; regenerating the split from the image group file reproduces exactly the sizes of the three sets recorded in `runs.csv`.

Augmentation: Each transformation consists of a random crop resized to 224×224 (area 0.6–1, aspect ratio 3/4–4/3), horizontal and vertical flips, a rotation drawn uniformly from $[-30^\circ, 30^\circ]$, and multiplication of brightness, contrast, and saturation by factors drawn uniformly from $[0.7, 1.3]$. The five offline augmented copies of each image have parameters that are a deterministic function of the seed, the image index, and the copy index, so they are identical in every run using the same seed, in both arms of Part B, and for both architectures; they are regenerated when the images are loaded rather than stored as files. Online augmentation is used only in RAND-SPLIT and GROUP-SPLIT: each time a training image is loaded, new parameters are drawn. Validation and test images are only center-cropped to 224×224 , except in AUG-SPLIT, where the validation and test sets also contain augmented copies.

Sampling per epoch: In Part A, each epoch iterates over the entire training set in a new shuffled order. In Part B, the training set of MATCH-CTRL consists of the original training images and five augmented copies of each image; MATCH-LEAK additionally contains five augmented copies of each test image. In each epoch, the sample list is shuffled and only the first $2N$ samples are used in both arms, where N is the number of original training images, so the two arms have the same number of update steps per epoch and the same maximum budget of 20 epochs; neither arm uses online augmentation. The validation and test sets are identical in the two arms and contain only center-cropped original images, so early stopping in MATCH-LEAK uses no leaked images. Because of early stopping on the validation set, the number of completed epochs (column `epochs_run` in `runs.csv`) differs between runs, so the total number of update steps of the two arms is not equal; Section 7 of the main paper reports this difference and an analysis by number of epochs.

Training and weight selection: The network is initialized from ImageNet weights, the classification head is replaced, and the whole network is fine-tuned; the optimization settings are given in Table 15 of the main paper. After each epoch, accuracy on the validation set is computed; training stops when this accuracy has not improved for five epochs or after 20 epochs, and the weights with the highest validation accuracy are restored. The test set and the PlantDoc [21] external test set (375 images of three disease classes, mapped to the PlantVillage class names) are evaluated only once with those weights. Seed s also initializes the classification head, the shuffling order, and the online augmentation. Because of fp16 mixed precision and non-deterministic GPU operations, rerunning with the same seed produces similar but not bit-identical results.

Aggregation and confidence intervals: Each cell of Table 14 is the mean and standard deviation over five seeds. In Part B, the difference $d_s = \text{MATCH-LEAK} - \text{MATCH-CTRL}$ is paired by seed; the 95% confidence interval is $\bar{d} \pm t_{0.975; n-1} s_d / \sqrt{n}$ with $n = 5$ and $t_{0.975; 4} \approx 2.776$. Differences between protocols in Part A are reported as differences of means.

What can and cannot be verified: From the data repository, readers can regenerate the set sizes from the image group files and check each numerical result in Tables 14 and 15 from the per-run results; the training

cannot be rerun without the code, which is available upon request.

References

- [1] Lucas M. Bachmann, Reto Coray, Pius Estermann, and Gerben ter Riet. Identifying diagnostic studies in MEDLINE: reducing the number needed to read. *Journal of the American Medical Informatics Association*, 9(6):653–658, 2002. doi: [10.1197/jamia.M1124](https://doi.org/10.1197/jamia.M1124).
- [2] Wichor M. Bramer, Melissa L. Rethlefsen, Jos Kleijnen, and Oscar H. Franco. Optimal database combinations for literature searches in systematic reviews: a prospective exploratory study. *Systematic Reviews*, 6:245, 2017. doi: [10.1186/s13643-017-0644-y](https://doi.org/10.1186/s13643-017-0644-y).
- [3] John Rathbone, Matt Carter, Tammy Hoffmann, and Paul Glasziou. Better duplicate detection for systematic reviewers: evaluation of systematic review assistant-deduplication module. *Systematic Reviews*, 4:6, 2015. doi: [10.1186/2046-4053-4-6](https://doi.org/10.1186/2046-4053-4-6).
- [4] smaranjitghose. Corn or Maize Leaf Disease Dataset. Kaggle, 2020. URL <https://www.kaggle.com/datasets/smaranjitghose/corn-or-maize-leaf-disease-dataset>. Kaggle dataset, version 1 (11 November 2020); compiled from PlantVillage and PlantDoc; accessed 3 October 2026.
- [5] Prabira Kumar Sethy. Rice leaf disease image samples. Mendeley Data, 2024. Version 2, published 18 July 2024.
- [6] Sayash Kapoor and Arvind Narayanan. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9):100804, 2023. doi: [10.1016/j.patter.2023.100804](https://doi.org/10.1016/j.patter.2023.100804).
- [7] A. Suksukont and E. Naowanich. Designing a squeeze-and-excitation-capsule BiLSTM-transformer for plant leaf disease recognition. *IAES International Journal of Artificial Intelligence*, 14(6):5069–5080, 2025. doi: [10.11591/ijai.v14.i6.pp5069-5080](https://doi.org/10.11591/ijai.v14.i6.pp5069-5080).
- [8] P. Tharani Pavithra and B. Baranidharan. OSPS-MicroNet: a distilled knowledge micro-CNN network for detecting rice diseases. *Frontiers in Computer Science*, 6:1279810, 2024. doi: [10.3389/fcomp.2024.1279810](https://doi.org/10.3389/fcomp.2024.1279810).
- [9] T.A. Prasetyo, V.L. Desrony, H.F. Panjaitan, R. Sianipar, and Y. Pratama. Corn plant disease classification based on leaf using residual networks-9 architecture. *International Journal of Electrical and Computer Engineering*, 13(3):2908–2920, 2023. doi: [10.11591/ijece.v13i3.pp2908-2920](https://doi.org/10.11591/ijece.v13i3.pp2908-2920).
- [10] J. Liu, Y. Hu, Q. Su, J. Guo, Z. Chen, and G. Liu. Semi-Supervised One-Stage Object Detection for Maize Leaf Disease. *Agriculture*, 14:1140, 2024. doi: [10.3390/agriculture14071140](https://doi.org/10.3390/agriculture14071140).
- [11] Ludo Waltman. A review of the literature on citation impact indicators. *Journal of Informetrics*, 10(2): 365–391, 2016. doi: [10.1016/j.joi.2016.02.007](https://doi.org/10.1016/j.joi.2016.02.007).
- [12] Romiyal George, Selvarajah Thuseethan, Roshan G. Ragel, Kayathiri Mahendrakumaran, Sivaraj Nimishan, Chathrie Wimalasooriya, and Mamoun Alazab. Past, present and future of deep plant leaf disease recognition: A survey. *Computers and Electronics in Agriculture*, 234:110128, 2025. doi: [10.1016/j.compag.2025.110128](https://doi.org/10.1016/j.compag.2025.110128).
- [13] Edwin B. Wilson. Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158):209–212, 1927. doi: [10.1080/01621459.1927.10502953](https://doi.org/10.1080/01621459.1927.10502953).
- [14] Lawrence D. Brown, T. Tony Cai, and Anirban DasGupta. Interval estimation for a binomial proportion. *Statistical Science*, 16(2):101–133, 2001. doi: [10.1214/ss/1009213286](https://doi.org/10.1214/ss/1009213286).
- [15] Rob J. Hyndman and Yanan Fan. Sample quantiles in statistical packages. *The American Statistician*, 50(4):361–365, 1996. doi: [10.1080/00031305.1996.10473566](https://doi.org/10.1080/00031305.1996.10473566).

- [16] Henry B. Mann and Donald R. Whitney. On a test of whether one of two random variables is stochastically larger than the other. *The Annals of Mathematical Statistics*, 18(1):50–60, 1947. doi: [10.1214/aoms/1177730491](https://doi.org/10.1214/aoms/1177730491).
- [17] Dave S. Kerby. The simple difference formula: An approach to teaching nonparametric correlation. *Comprehensive Psychology*, 3:11.IT.3.1, 2014. doi: [10.2466/11.IT.3.1](https://doi.org/10.2466/11.IT.3.1).
- [18] Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979. URL <https://www.jstor.org/stable/4615733>.
- [19] David P. Hughes and Marcel Salathé. An open access repository of images on plant health to enable the development of mobile disease diagnostics. *arXiv preprint arXiv:1511.08060*, 2015. doi: [10.48550/arXiv.1511.08060](https://doi.org/10.48550/arXiv.1511.08060).
- [20] A. Petchiammal, S. Briskline Kiruba, D. Murugan, and Pandarasamy Arjunan. Paddy Doctor: A visual image dataset for automated paddy disease classification and benchmarking. In *Proceedings of the 6th Joint International Conference on Data Science & Management of Data (CODS-COMAD)*, pages 203–207, 2023. doi: [10.1145/3570991.3570994](https://doi.org/10.1145/3570991.3570994).
- [21] Davinder Singh, Naman Jain, Pranjali Jain, Pratik Kayal, Sudhakar Kumawat, and Nipun Batra. PlantDoc: a dataset for visual plant disease detection. In *Proceedings of the 7th ACM IKDD CoDS and 25th COMAD*, pages 249–253, 2020. doi: [10.1145/3371158.3371196](https://doi.org/10.1145/3371158.3371196).